Algebra 1

FINAL EXAM

2022 / 2023

Duration : 03 hours

Exercise 01 Prove or disprove the following statements.

1. For $f : \mathbb{R} \longrightarrow \mathbb{R}$ a map defined by $f(x) = x^2$, we have $f^{-1}(f([-1, 1])) = [-1, 1]$.
2. On $\mathbb{N} \times \mathbb{N}$, the binary relation R_1 defined by : $\forall (a, b), (c, d) \in \mathbb{N} \times \mathbb{N}$, $(a, b)R_1(c, d)$ if and only if $a \leq c$ is antisymmetric.
3. On $\mathbb{R}$, the binary relation R_2 defined by : $\forall x, y \in \mathbb{R}$, xR_2y if and only if $x \leq 2^y$ is neither symmetric nor antisymmetric.
4. For any a, b, and c in an integral domain, if $a \neq 0$ and $ab = ac$ then $b = c$.
5. Every subring is an ideal.
6. If $\varphi : R \longrightarrow R'$ is a ring homomorphism, then $\varphi(-r) = -\varphi(r)$ for any $r \in R$.
7. In a commutative ring with unity an element $x \neq 0$ is either invertible or zero divisor.
8. Let $P \in \mathbb{R}[X]$ and $z_0 \in \mathbb{C}$. If $P(z_0) = 0$ then $P(\overline{z_0}) = 0$.
9. $X^4 + 1$ is irreducible in $\mathbb{R}[X]$.
10. $X^3 - 2$ is irreducible in $\mathbb{Q}[X]$.

Exercise 02 Let n be a nonnegative integer and define the polynomial

$$P_n(X) := X^6 + (n-1)X^5 + 3X^4 + 2(n-1)X^3 + 11X^2 + 9(n-1)X + 9.$$

1. Find the remainder of the Euclidean division of $X^{n+2} + P_n(X)$ by $X^3 - X^2$.
2. Show that $P_0(X)$ and $X^2 - 1$ are coprime, then find two polynomials $U(X)$ and $V(X)$ such that

$$U(X)P_0(X) + V(X)(X^2 - 1) = 1.$$

3. (a) Show that i is a root of $P_n(X)$ if and only if $n = 1$ (i is the complex number which satisfies $i^2 = -1$).
   (b) Write $P_1(X)$ as a product of irreducible factors in $\mathbb{R}[X]$ and $\mathbb{C}[X]$.
   (c) Split in $\mathbb{R}(X)$ the rational fraction

$$\frac{2(X-1)^2(X^2+2X+3)}{P_1(X)}.$$

Exercise 03

1. Show that any group of prime order is cyclic.
2. Let $n \geq 2$ be an integer. Let $1 \leq k \leq n$ and

$$\begin{array}{rccc} \varphi_k : & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & \mathbb{Z}/n\mathbb{Z} \\ & \overline{x} & \longmapsto & \overline{kx}. \end{array}$$

   (a) Show that φ_k is a group endomorphism.

   (b) Let $\varphi : \mathbb{Z}/n\mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z}$ be a map. Show that φ is a group endomorphism if and only if there exists an integer $k \in \{1, \ldots, n\}$ such that $\varphi = \varphi_k$.

   (c) Show that φ_k is a ring endomorphism if and only if n divides $k(k-1)$.

   (d) What are the ring endomorphisms of $\mathbb{Z}/4\mathbb{Z}$?

   (e) What are the ring endomorphisms of $\mathbb{Z}/6\mathbb{Z}$?

3. Let $G := \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. We define on G the addition $\oplus$ as :
$\forall (\overline{a}, \overline{b}), (\overline{c}, \overline{d}) \in G : (\overline{a}, \overline{b}) \oplus (\overline{c}, \overline{d}) = (\overline{a+c}, \overline{b+d})$.

   (a) What is the cardinality of G? List the elements of G and draw the addition table of G.

   (b) Can we say that all the proper subgroups of G are cyclic ?

   (c) Give all the proper subgroups of $(G, \oplus)$.

   (d) Justify why $(G, \oplus)$ is not isomorphic to $(\mathbb{Z}/4\mathbb{Z}, +)$.

   (e) Let H be a group such that all its proper subgroups are cyclic. Can we say that H is cyclic ?

4. Now we define on G the multiplication $\odot$ as : $\forall (\overline{a}, \overline{b}), (\overline{c}, \overline{d}) \in G : (\overline{a}, \overline{b}) \odot (\overline{c}, \overline{d}) = (\overline{ac}, \overline{bd})$.

   (a) Draw the multiplication table of G.

   (b) Knowing that $(G, \oplus, \odot)$ is a commutative ring with $(\overline{1}, \overline{1})$ as unity, determine :

      i. D the set of the zero divisors.

      ii. U the set of invertible elements.

      iii. N the set of nilpotent elements.

      iv. I the set of idempotent elements.

   (c) What is the characteristic of G?

   (d) Is $(G, \oplus, \odot)$ a field ?

5. Let $n, m \geq 2$ be two integers. In this part, $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ is equipped with the addition $\boxplus$ defined by : $\forall (\overline{a}, \widehat{b}), (\overline{c}, \widehat{d}) \in \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} : (\overline{a}, \widehat{b}) \boxplus (\overline{c}, \widehat{d}) = (\overline{a+c}, \widehat{b+d})$.

   (a) Show that the order of $(\overline{1}, \widehat{1})$ in $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is $\operatorname{lcm}(n, m)$.

   (b) Deduce that $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is isomorphic to $(\mathbb{Z}/nm\mathbb{Z}, +)$ if and only if $gcd(n, m) = 1$.

Good luck